RAJALAKSHMI ENGINEERING COLLEGE

Mahalakshmi B 250701401-2025-CSE

M2

Started on	Friday, 25 September 2026, 2:25 PM
State	Finished
Completed on	Friday, 25 September 2026, 2:26 PM
Time taken	35 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int findDuplicate(const int* arr, int n) {
    // Phase 1: Find the intersection point of the two runners
    int tortoise = arr[0];
    int hare = arr[0];

    do {
        tortoise = arr[tortoise];
        hare = arr[arr[hare]];
    } while (tortoise != hare);

    // Phase 2: Find the entrance to the cycle (the duplicate element)
    tortoise = arr[0];
    while (tortoise != hare) {
        tortoise = arr[tortoise];
        hare = arr[hare];
    }

    return tortoise;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int* arr = (int*)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &arr[i]) != 1) {
            free(arr);
            return 0;
        }
    }

    int duplicate = findDuplicate(arr, n);
    printf("%d\n", duplicate);

    free(arr);
    return 0;
}

```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔

	Input	Expected	Got	
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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